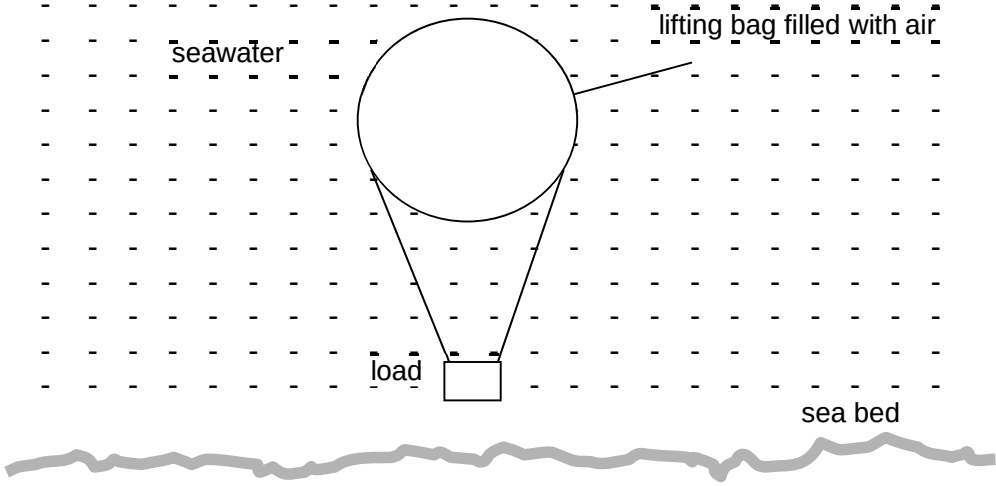


2	(a)	Explain the origin of upthrust acting on a body in a fluid.	
			
			[2]
		Fluid pressure increases (directly proportional) with depth in the fluid. This pressure differences results in a net upward force exerted by the fluid on the body, which is known as upthrust.	1 1
	(b)	In order to lift a submerged load of 600 kg from a seabed, a lifting bag made of an elastic material of negligible mass is filled with air and attached to the load, as shown in Fig. 2.1. The density of seawater is 1050 kg m^{-3} and the lifting bag contains 0.700 m^3 of air of density 1.27 kg m^{-3} such that the load ascends with a constant speed. The volume of the load is negligible compared to the volume of the lifting bag.  Fig. 2.1 (not to scale)	
	(i)	Calculate the upthrust on the lifting bag when its volume is 0.700 m^3 .	
		Upthrust = Weight of seawater displaced = density $\times$ volume $\times$ g = $1050 \times 0.700 \times 9.81$ = $7210.35 = 7210 \text{ N}$	1 1
		upthrust = N	[2]

			(ii) Calculate the total drag force on the bag and load.	[2]
			$U = mg + F_D$ $F_D = 7210.35 - 600 \times 9.81$ $= 1.32 \times 10^3 \text{ N}$	1 1
			total drag force = N	[2]
			(iii) Explain why in practice, to maintain a constant speed of ascent, air has to be released continuously from the lifting bag. Temperature change during the ascent is negligible.	
				
				[2]
			Net force must remain equal to zero.	
			Since the drag force and weight constant, upthrust must be constant and hence the volume of the lifting bag must be constant.	1
			As fluid pressure decreases as the bag ascends, the volume of air in the bag increases. Therefore, air has to be let out so as to maintain constant volume of the bag.	1